Amplitude Magnetic Permeability

https://github.com/DmitriyMakhnovskiy/Tellinen-Madelung-Model-BHloops/tree/main/Amplitude-Permeability-Tellinen-Madelung-Model

In the technical description of Fair-Rite 61 Material™, which we regard as a benchmark for product technical documentation, an interesting experimental characteristic is presented - the dependence of amplitude magnetic permeability μ_a on the amplitude of magnetic flux density B. It is useful to examine this question if, in the future, we intend to measure hysteresis loops ourselves.

Consider an arbitrary cyclic symmetric minor loop nested within the major loop. Let the maximum values of magnetic field strength and magnetic flux density on this loop be denoted by H_{max} and B_{max}, respectively. The relative amplitude magnetic permeability is defined by the following ratio (in SI units): $\mu_a = \frac{B_{max}}{\mu_0 H_{max}}$. For a family of symmetric minor loops with different excitation amplitudes, each pair of values $(H_{max,i}, B_{max,i})$ corresponds to a value $\mu_{a,i}$. The resulting set of points makes it possible to construct the dependence $\mu_a(B)$.

To determine the complete dependence of amplitude magnetic permeability $\mu_a(B)$, measurements should be performed starting from a previously demagnetized state of the material and gradually increasing the amplitude of the periodic scanning magnetic field $H(t)$. During experimental determination of the sequence of value pairs $(H_{max,i}, B_{max,i})$, the increase in scanning-field amplitude may be periodically paused while cyclic remagnetization is maintained at a fixed amplitude, and then resumed. These pauses make it possible to record the corresponding minor hysteresis loops and determine their amplitude characteristics. It is essential that, in this case, the magnetization remains cyclic throughout the entire measurement process. This method therefore differs fundamentally from measurement of the initial magnetization curve $B_{init}(H)$, in which the magnetic field $H(t)$ increases monotonically from zero, forming the primary magnetization trajectory of the previously demagnetized material. From the amplitude-permeability measurement results, one can construct the locus of the positive vertices of the minor hysteresis loops, defined by the sequence of points $(H_{max,i}, B_{max,i}) = \left(\frac{B_{max,i}}{\mu_0 \mu_a(B_{max,i})}, B_{max,i}\right)$. However, the curve obtained in this way does not, in general, have to coincide with the initial magnetization curve $B_{init}(H)$. The reason is that different magnetic states of the material are formed under monotonic and cyclic field variation as a result of hysteresis and magnetic memory. Consequently, reconstructing the initial magnetization curve from the experimental dependence $\mu_a(B)$ requires additional justification and cannot be regarded as a universally applicable method.

For Fair-Rite 61 Material™, the experimental dependences $\mu_a(B)$ for two temperatures are shown in Fig. 1; the minor loops were measured at a frequency of 10 kHz.

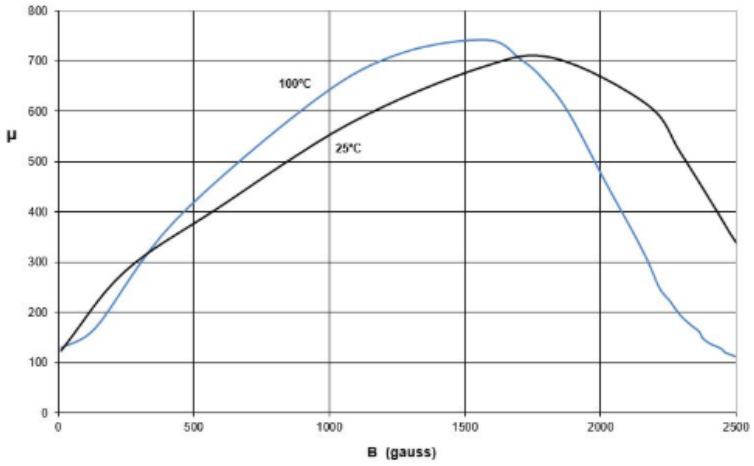


Fig. 1. Dependence of amplitude magnetic permeability on magnetic flux density for Fair-Rite 61 Material™ in cgs units.

Note that the graph in Fig. 1 is not fully consistent with the major hysteresis loop from the same datasheet. A flux density of $B = 2500$ G, which is close to saturation, corresponds on the major loop to a field of $H \approx 13$ Oe; therefore, we would expect $\mu_a = \frac{2500}{13} \approx 192.31$. However, the graph shows a value of ≈ 350 at 25 °C. The datasheet itself provides no explanation of these graphs. The conditions under which the minor loops used to calculate $\mu_a(B)$ were measured therefore need to be clarified.

The dependence of amplitude magnetic permeability on magnetic flux density shows what magnitude of magnetic field is required to reach a specified flux density under symmetric cyclic remagnetisation. Each point on this dependence corresponds to the vertex of one symmetric hysteresis loop. Therefore, the entire curve $\mu_a(B)$ contains information about the locations of loop vertices at different excitation amplitudes. This characteristic makes it possible to check how correctly the model describes remagnetization inside the major loop. Two models may reproduce the major loop equally well but yield different minor-loop vertices and, consequently, different amplitude permeability. At the same time, the positions of the vertices do not determine the shapes of the loops: their widths, remanent flux densities, and areas may differ even when the vertices coincide. Agreement with the experimental amplitude-permeability dependence is therefore an important, but not complete, validation of the minor-loop model.

Once the major loop has been specified, the Chan–Engelhardt and Tellinen–Madelung models do not use information about minor loops; instead, they generate them according to internal algorithmic rules that are physically motivated but nevertheless model-based. In other words, the experimental data primarily determine the material's major loop, whereas the minor loops in both models are a consequence of the selected mechanism used to construct them. More sophisticated models trained on minor-loop measurements also exist, but we do not consider them in this report because such detail is not significant for our applications. Nevertheless, for the purpose of validating the Tellinen–Madelung model, it would be interesting to calculate $\mu_a(B)$ and compare it with Fig. 1. The Python algorithm for calculating $\mu_a(B)$ within the Tellinen–Madelung model is stored in the project folder "Amplitude-Permeability-Tellinen-Madelung-Model" in the GitHub repository. As the major loop, we use the digitized Fair-Rite 61 Material™ loop at 25 °C. Ideally, $B_{init}(H)$ should be provided together with the

major-loop branches. Since it was absent from the datasheet, we calculated it as the mean curve of the major-loop branches, similarly to the procedure used in the Chan–Engelhardt model. Because the Tellinen–Madelung model includes magnetic memory, we reproduced the experimental excitation conditions in which the sample is first demagnetized by a periodic field of decreasing amplitude and is then remagnetized to a specified level by a periodic field of increasing amplitude. The following functions are used for this purpose in the Python algorithm (file Hscan.py), where the parameter Hb (shown later in the figures) defines the limiting amplitude of the magnetizing field:

```
def demag_amplitude(t):
    """Exponential fall from Hmajor towards the small positive floor."""
    t = np.asarray(t, dtype=float)
    return Hfloor + (Hmajor - Hfloor) * np.exp(-t / (3.0 * T))


# Exact envelope at the switch, used in both stages (no field jump).
Hswitch = float(demag_amplitude(t_demag))


def Hdemag(t):
    """First stage: alternating demagnetization from +Hmajor."""
    t = np.asarray(t, dtype=float)
    return demag_amplitude(t) * np.cos(w * t)


def Hscan(t, Hb):
    """Second stage, absolute time t >= t_demag: growing AC field.

    H(t_demag) = Hswitch; amplitude approaches Hb from below.
    The oscillation phase is inherited from Hdemag, without a jump.
    """
    t = np.asarray(t, dtype=float)
    tau = t - t_demag
    if np.any(tau < -1e-12):
        raise ValueError("Hscan requires t >= t_demag")
    if Hb <= Hswitch:
        raise ValueError(f"Hb ({Hb:g}) must exceed Hswitch ({Hswitch:g}) A/m")
    amplitude = Hswitch + (Hb - Hswitch) * (1.0 - np.exp(-tau / (3.0 * T)))
    return amplitude * np.cos(w * t)
```

An example of the combined excitation and the accompanying minor-loop dynamics is shown in Fig. 2 for Hb = 200 A/m, i.e., before magnetic saturation is reached. The demagnetization process starts from some large field Hstart, for example from a saturated state. In Fig. 2, the starting field was 1000 A/m. The field then begins to vary cyclically with decreasing amplitude, causing the minor loops to contract toward zero. At the end of this process, a demagnetized state is obtained in which the magnetization averaged over the sample volume is approximately zero (although not locally within the magnetic domains). This is followed by magnetization in a cyclic field whose amplitude increases to the next limiting value Hb. Once the field amplitude reaches its limiting value, a symmetric minor loop is obtained. On the final magnetization cycle, the dimensional parameters of the minor loop are determined, where $B_{max,i}$ corresponds to $H_{max,i} \approx \mathrm{Hb}$. These two values are then used to calculate $\mu_{a,i}$. The amplitude of the cyclic magnetization can then be smoothly increased to move to the next minor loop. In the Python algorithm, the complete family of points $(H_{max,i}, B_{max,i})$ is obtained for a monotonically increasing sequence of Hb values, with each magnetization process starting from the same initial demagnetized state, which is calculated only once. Reconstruction of the complete curve $\mu_a(H)$ ends when Hb reaches its maximum value, as shown in Fig. 3.

In the numerical procedure described above for determining $\mu_a(H)$, we reproduce the conditions of a real experiment. Note that the vertices of the minor loops in the Tellinen–Madelung model do not lie on the curve $B_{init}(H)$; therefore, these are substantially different magnetic characteristics that can be obtained experimentally.

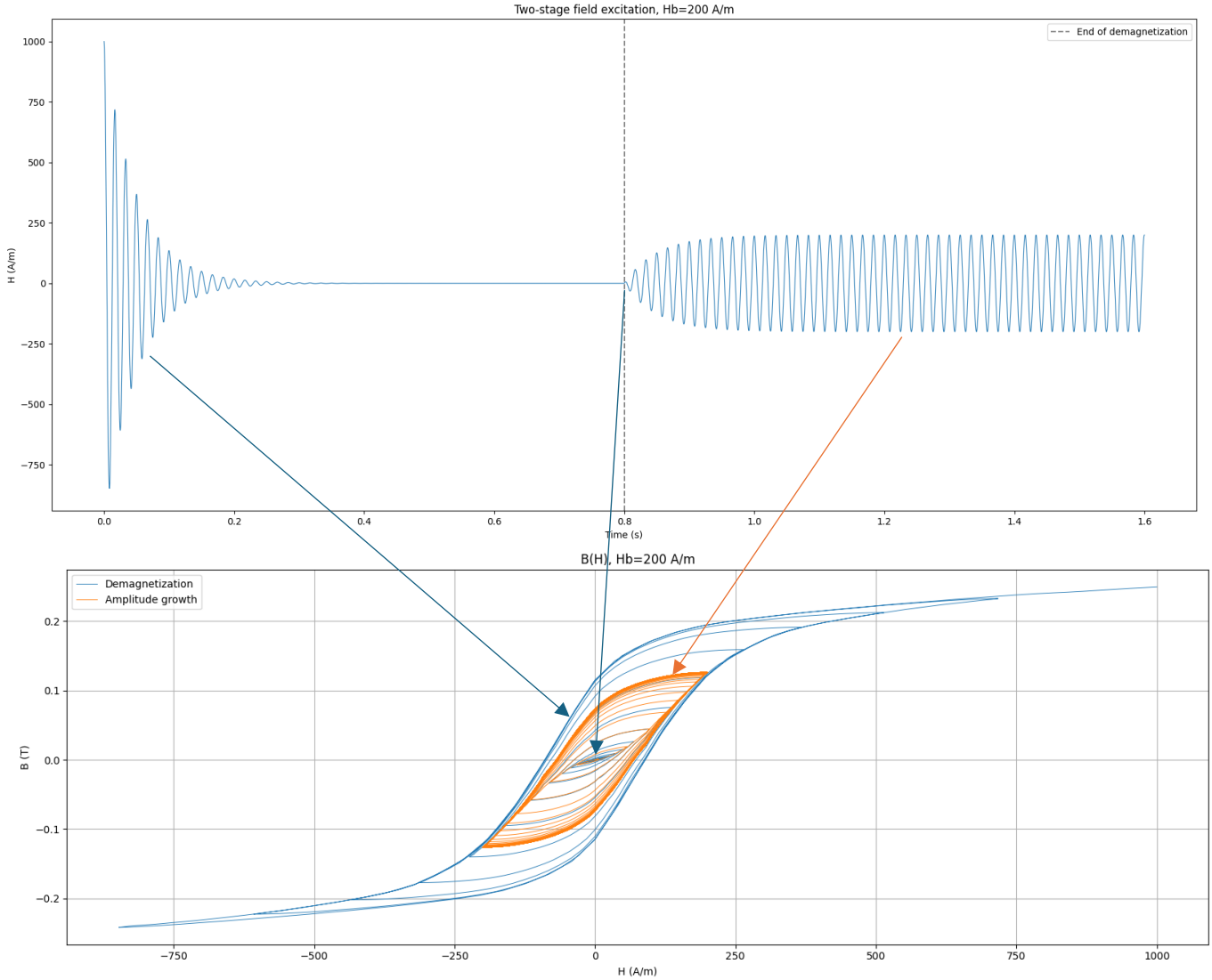


Fig. 2. Combined excitation $H(t)$, including cyclic demagnetization and subsequent magnetization, calculated for Fair-Rite 61 Material™ within the Tellinen–Madelung model. In the first stage, the minor loop contracts to a point (blue curves). This is followed by cyclic magnetization with increasing amplitude up to Hb = 200 A/m, which forms a stable symmetric minor loop (brown curves).

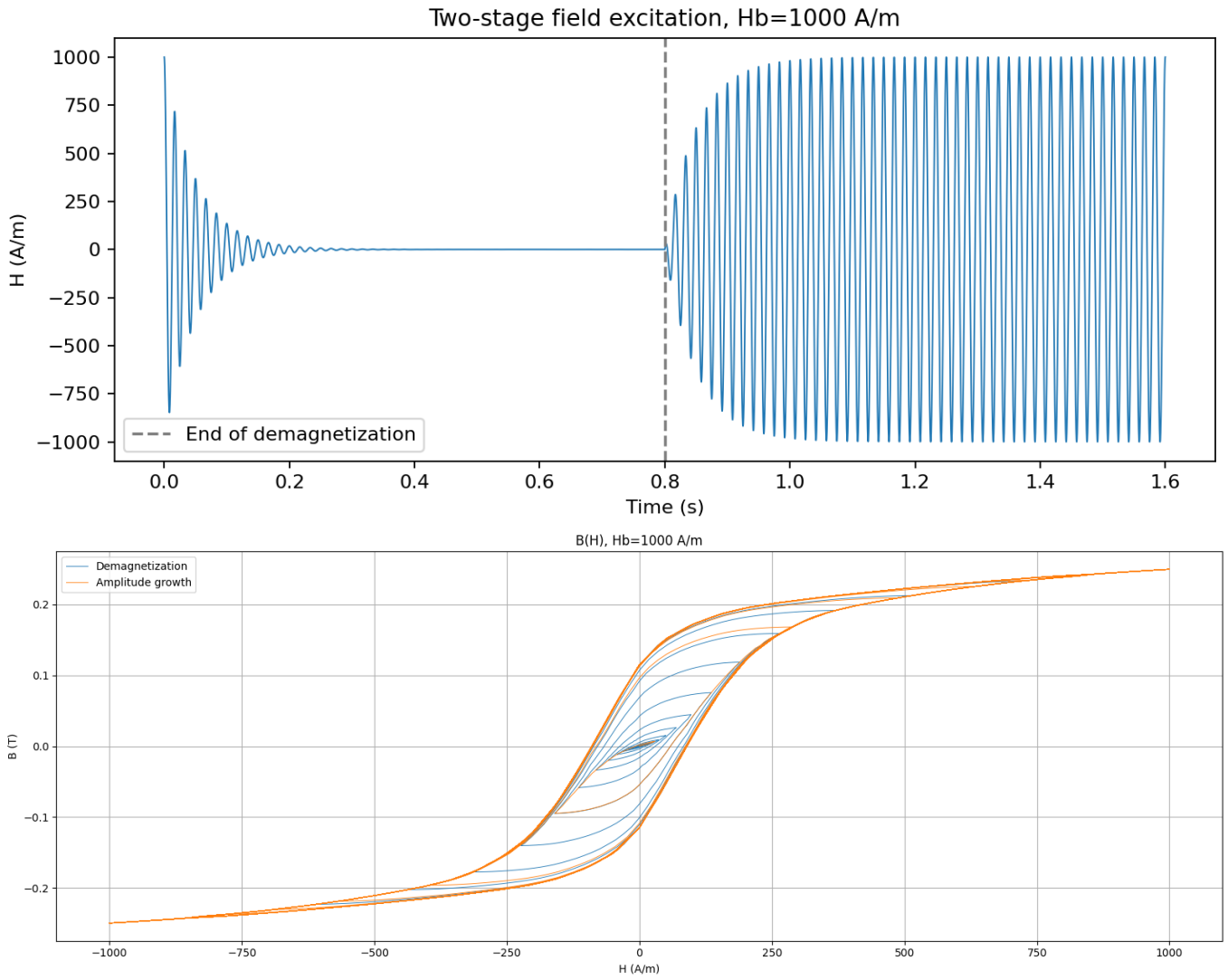


Fig. 3. Limiting minor loop coinciding with the original major loop, obtained during cyclic magnetization with increasing field amplitude.

Calculation parameters in the Parameters.txt file of the Python project:

```
# Two-stage Tellinen-Madelung amplitude-permeability experiment.
# Hmajor is the initial point H(0)=+Hmajor on the UPPER MAJOR branch.
# Choose it near positive saturation, inside the common measured H range.
Hmajor=1000.0

# First stage: sinusoidal demagnetization to the residual amplitude Hswitch.
# Hfloor is the asymptotic envelope floor (A/m), NOT a B-offset.
Hfloor=0.1
DemagPeriods=48

# Second stage: amplitude rises from Hswitch to the target Hb.
HbMin=10.0
HbMax=1000.0
HbPoints=30
RemagPeriods=48
SamplesPerPeriod=120

# Numerical tolerances and original Tellinen-Madelung integration options.
H_tol=1.0e-9
B_tol=1.0e-9
D_tol=1.0e-12
IntegrationMethod=lagrange
IntegrationIntervals=500
```

The resulting dependence of amplitude magnetic permeability $\mu_a(B)$, calculated using the procedure described above, is shown in Fig. 4. Quantitatively, it differs noticeably from the experimental dependence in Fig. 1: both the absolute level and individual features of the curve shape are different. At the same time, the overall character of the permeability variation with increasing magnetic flux density remains similar in both cases. However, a truly meaningful comparison of these dependences is limited by the lack of information about the method used for the experimental determination of the minor loops. Without knowing the excitation conditions, frequency, amplitude, and data-processing procedure, it is difficult to interpret the observed quantitative discrepancies unambiguously. In addition, as noted above, differences in the detailed shape of the minor loops appear to have only a weak effect on the current response of the inductor. Therefore, from the standpoint of circuit simulation, such discrepancies may be significantly less important than an accurate reproduction of the major loop, saturation, and the principal nonlinear properties of the magnetic material.

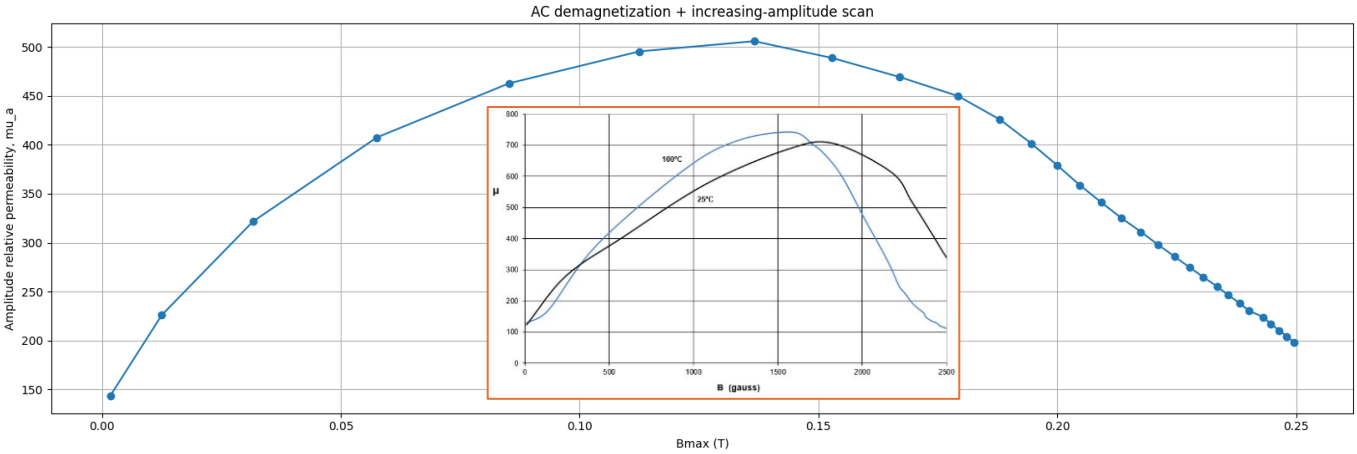


Fig. 4. Calculated dependence of amplitude magnetic permeability on magnetic flux density within the Tellinen–Madelung model, using the digitized major loop of Fair-Rite 61 Material™ at 25 °C. The experimental curves are inset for comparison.

The comparisons presented above show that, in circuit-simulation problems, the detailed implementation of magnetic memory may have a much smaller effect on the electrical response than an accurate description of the principal nonlinear properties of the magnetic circuit: saturation, coercivity, remanent flux density, and physically plausible behaviour of the minor loops. Therefore, for many electronics and electrical-engineering applications, it appears unnecessary to pursue the most complex and formally sophisticated hysteresis models if a simpler model adequately reproduces the observed currents and voltages. This is particularly important for calculations of the magnetic rapping actuator. It means that the result may be sufficiently robust with respect to a number of simplifying assumptions in the description of the magnetic circuit, including the use of approximate hysteresis models and a limited set of material parameters. In this case, attention should primarily be focused on the factors that directly determine the force and transient response of the actuator: magnetic-circuit geometry, the magnitude and variation of air gaps, saturation of individual core regions, winding resistance and parameters, and correct representation of the external electrical circuit.

In the Chan–Engelhardt model, the branches of the major loop are specified by analytical template functions. Such functions can reproduce the shape of the loop of a real material quite accurately. However, such agreement cannot be guaranteed in advance for every soft magnetic material. A more universal version of the model could be obtained by replacing the analytical template functions with experimental dependences $B(H)$, and then performing subsequent minor-loop calculations using their interpolated representations. This is exactly the approach we use in calculations with the Tellinen–Madelung model. Such a generalization would make the Chan–Engelhardt model more realistic in describing the magnetic properties of a particular material, especially the shape of its major loop. At the same time, the model would retain its important advantage: it would remain substantially less computationally expensive than Tellinen–Madelung because it does not require such a detailed description of magnetic memory, the nesting of preceding states, or numerical integration.

The differences between Chan–Engelhardt and Tellinen–Madelung are manifested mainly in the way magnetic memory and the nesting of minor loops are described, but they are not necessarily significant for calculating inductor current. This practical insensitivity of the result to details of the hysteresis model is of fundamental importance for engineering calculations. It provides grounds for expecting the computational models to remain predictive even when the material description is inevitably simplified. Otherwise, it would be difficult to speak of the reproducibility of calculations and design solutions for large electromagnetic devices such as transformers or plunger actuators if their characteristics depended strongly on subtle features of magnetic memory or on the particular scenario used to construct minor loops. For engineering modelling, it is important that the result be determined primarily by the main material characteristics, device geometry, and external-circuit parameters rather than by sensitive details of the selected hysteresis model. For this reason, the similarity of current responses obtained using different magnetic hysteresis models should be regarded as a positive result. It shows that, in the regime considered, the inductor calculation has a certain robustness with respect to model assumptions and can therefore be used as a practical tool for further analysis of electromagnetic devices, including the magnetic rapping actuator.